

Solar Energy Basics

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Solar panels convert sunlight into electricity using photovoltaic cells. The energy generated can be used immediately by your home appliances, or fed back into the grid for credits (net metering).

During peak sunlight hours, usually between 10 AM and 2 PM, solar generation is at its highest. A solar surplus occurs when your panels generate more power than your home is currently consuming. Utilizing high-power appliances like washing machines during solar surplus hours maximizes your savings.

Advanced Solar Technologies:

Beyond basic photovoltaic cells, modern solar systems incorporate advanced technologies to improve yield. Monocrystalline panels offer higher efficiency in smaller spaces compared to polycrystalline variants. Microinverters and power optimizers attach to individual panels, ensuring that shading on one panel doesn't pull down the production of the entire array. Furthermore, smart solar tracking systems can physically tilt the panels throughout the day to follow the sun's trajectory, increasing overall energy capture by up to 25%.

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